

## Education

**Dayananda Sagar College of Engineering**

**Bengaluru, India**

*B.E. Artificial Intelligence and Machine Learning*

## Experience

### **Bentham AI**

*Dec 2025 – Feb 2026*

*Backend Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions*

*Remote*

- Built backend automation around public-site API lookups, browser execution, retries, **session recovery**, validation checks, and operator checkpoints, reducing manual filing time by **85%**.
- Structured API responses, screenshots, generated drafts, form state, and failure checkpoints as **debuggable workflow records** for unreliable external systems.
- Containerized Node.js/Go services with **Docker**, deployed on **Google Cloud Run**, and wired **GitHub Actions** release workflows, improving deployment velocity by **70%**.
- Kept generated drafts and workflow outputs inspectable before final execution, matching regulated-product needs around traceability, review, and rollback.

### **Google Summer of Code**

*May 2025 – Sept 2025*

*Mentee at Invesalio: Python, VTK, wxPython, NumPy, PyTorch, ONNX*

*Remote*

- Integrated **Python workflows** for **95 anatomical brain regions**, wiring preprocessing, PyTorch/TorchScript and ONNX/TinyGrad execution paths, generated artifacts, and correctness checks into Invesalio.
- Built model-output inspection paths around generated masks, label remapping, orientation checks, progress reporting, and reproducible artifact debugging.
- Reduced expensive processing runtime by **87%** through async/parallel execution while debugging GPU/CPU paths, large data movement, and output correctness.

## Open Source

*Contributions* 

- Contributed maintainer-reviewed patches across **Kuadrant MCP Gateway**, **Kyverno**, **Invesalio**, **Tiled**, **CRIU**, and **VideoLAN**, spanning backend failover behavior, parser robustness, security validation, compatibility fixes, and data/artifact correctness.

## Projects

**Postificus Content Distribution Engine** | *Go, Echo, React, PostgreSQL, Redis/RabbitMQ, Go-Rod*

**GitHub** 

- Built a centralized **content distribution engine** for writing once and publishing technical posts across Medium, Dev.to, and LinkedIn while preserving canonical URL/SEO ownership.
- Split the Go/Echo backend into API and worker paths, using PostgreSQL persistence, Redis caching, RabbitMQ queues with retry/DLQ handling, and Prometheus metrics.
- Implemented direct platform API flows where available and **Go-Rod browser automation** as fallback for platforms without reliable write APIs.

**Hyoka Agent Reliability Layer** | *Python, FastAPI, SQLAlchemy, PostgreSQL, OpenTelemetry*

**GitHub** 

- Built the production core for an **agent reliability layer**: trace ingestion, eval/replay runs, failure mining, candidate generation, gate decisions, promotions, artifacts, manifests, and audit logs.
- Designed **SQLite/PostgreSQL metadata models** for traces, suites, candidates, runs, artifacts, manifests, worker leases, gate decisions, and promotions using SQLAlchemy/Alembic.
- Implemented database-backed worker leases, content-addressed artifacts, HMAC-signed manifests, API-key isolation, and Docker Compose services for server, worker, proxy, dashboard, and Postgres.

**Seaweed Contest Platform** | *Go, Next.js, PostgreSQL, Firebase Auth, S3, Judge0*

**GitHub** 

- Built a **coding contest platform** for live rounds, problem publishing, Firebase-authenticated registrations, submissions, admin controls, and public leaderboards.
- Implemented Go/Echo APIs over PostgreSQL for contests, problems, submissions, test-case results, rankings, admin flags, and materialized leaderboard refreshes.
- Integrated Judge0 judging, S3 submission storage, Next.js problem/leaderboard/admin pages, and score/attempt aggregation for **500+ concurrent users**.

## Technical Skills

**Languages:** Go, Python, TypeScript, JavaScript

**Frameworks:** FastAPI, Node.js, React, Echo, Gin, SQLAlchemy

**Tools:** Docker, Kubernetes, GCP Cloud Run, GitHub Actions, PostgreSQL, Redis, RabbitMQ, Prometheus, OpenTelemetry, REST APIs